

Band offsets of wide-band-gap oxides and implications for future electronic devices

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Wide-band-gap oxides such as SrTiO_3 are shown to be critical tests of theories of Schottky barrier heights based on metal-induced gap states and charge neutrality levels. This theory is reviewed and used to calculate the Schottky barrier heights and band offsets for many important high dielectric constant oxides on Pt and Si. Good agreement with experiment is found for barrier heights. The band offsets for electrons on Si are found to be small for many key oxides such as SrTiO_3 and Ta_2O_5 which limit their utility as gate oxides in future silicon field effect transistors. The calculations are extended to screen other proposed oxides such as BaZrO_3 , ZrO_2 , HfO_2 , La_2O_3 , Y_2O_3 , HfSiO_4 , and ZrSiO_4 . Predictions are also given for barrier heights of the ferroelectric oxides $\text{Pb}_{1-x}\text{Zr}_x\text{TiO}_3$ and $\text{SrBi}_2\text{Ta}_2\text{O}_9$ which are used in nonvolatile memories. © 2000 American Vacuum Society. [S0734-211X(00)10503-7]

I. INTRODUCTION

The theory of Schottky barrier heights between a metal and a semiconductor and of band alignments at heterojunctions between two semiconductors has been intensely debated.^{1–8} These theories were usually tested on zinc-blende semiconductors, because of the greater work on their behavior and band structure. In addition, the band offsets in these materials are critical to their use in heterojunction lasers and in high mobility transistors. The wide-band-gap oxides provide another interesting test of the theories. Furthermore, the predictions of the models will be shown to be critical to applications of such oxides in future microelectronics.

There is increasing interest in microelectronics for oxides with a higher dielectric constant K such as tantalum pentoxide and barium strontium titanate (BST) for use in integrated capacitors, as the dielectric layer in dynamic random access memories (DRAMs).^{9–19} There is also strong interest in ferroelectrics such as lead zirconate titanate (PZT) and strontium bismuth tantalate (SBT) for use in nonvolatile memories.^{20–22} In each case, leakage current is an important issue in device performance.

There is also an urgent need to find alternative dielectrics to act as the gate insulator in metal–oxide–silicon technology. Silicon dioxide is presently used, but the oxide thickness will soon be so small that its leakage will be too high for device reliability.²³ The alternative is to use larger thickness of a higher dielectric constant insulator. However, these new oxides such as BST have much lower gaps than silicon dioxide and they could have a different class of leakage problems due to Schottky emission of carriers into the band states.

The requirements for these oxides are severe.^{24–26} They must have a high thermodynamic stability so that they do not react with silicon. Their diffusion coefficients should be low so that they can withstand processing temperatures. For gate oxides, they should form high quality interfaces with silicon with few interfacial defects and low roughness. The fourth

requirement is that they have barriers for electrons and holes of at least 1 eV, in order to have sufficiently low leakage currents due to Schottky emission. The importance of Schottky emission leakage in these applications can be found by treating the oxides as wide-band-gap semiconductors and finding their band alignments with silicon.

II. SCHOTTKY BARRIERS

When two metals of different work functions approach each other, there is some charge transfer between them to give an external voltage, which equalises their Fermi levels. At a metal–semiconductor contact, there may or may not be charge transfer. For no charge transfer, in the Schottky limit, the barrier height for electrons ϕ_n is given by the difference between the work function of the metal Φ_m and the electron affinity (EA) of the semiconductor, χ_s ,

$$\phi_n = \Phi_m - \chi_s. \quad (1)$$

Generally though, there is some charge transfer between the metal and interface states lying in the semiconductor band gap, as shown in Fig. 1. The charge transfer creates a dipole, which reduces the electron Schottky barrier height (SBH) to

$$\phi_n = S(\Phi_m - \Phi_s) + (\Phi_s - \chi_s). \quad (2)$$

Here, Φ_s is the energy of the semiconductor interface states below the vacuum level, S is the Schottky pinning parameter, which is given in the linear approximation by²⁷

$$S = \frac{1}{1 + (e^2 N \delta) / (\epsilon \epsilon_0)} \quad (3)$$

and e is the electronic charge, ϵ_0 is the permittivity of free space, N is the density of the interface states per unit area, and δ is their extent into the semiconductor. The slope parameter S is characteristic of the semiconductor, with $S=1$ describing the Schottky limit of no pinning in which the SBH is given by Eq. (1) and $S=0$ describing the Bardeen

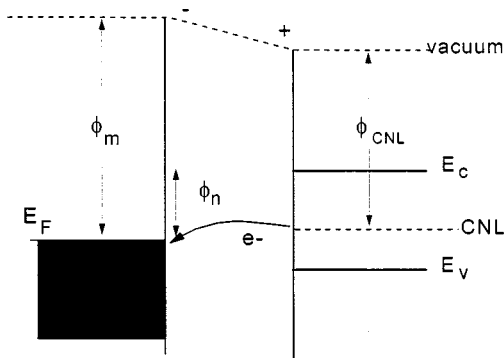


FIG. 1. Schematic of the effect of charge transfer between semiconductor midgap states and the metal on the Schottky barrier height.

limit of strong pinning. In the strong pinning limit, the metal Fermi level is pinned by the interface states at Φ_S and the SBH is $\Phi_S - \chi_s$.

There are two origins of interface gap states, intrinsic and extrinsic. In the intrinsic model of Bardeen and Heine,²⁸ a semi-infinite semiconductor in contact with a metal possesses intrinsic states within the energy gap of the semiconductor. These states are sometimes called “metal-induced gap states” (MIGS).²⁹ They can be thought of as the dangling bond states of the broken surface bonds dispersed across the band gap of the semiconductor, or as the evanescent states of the metal wave functions continued into the forbidden energy gap of the semiconductor.²⁸ Φ_S is then the “charge neutrality level” (CNL) of the interface states, defined as the energy above which the states are empty for a neutral surface.^{29–32}

The interface between a semiconductor and a metal is often accompanied by a chemical reaction. The Schottky barrier height has been correlated to the degree of this reaction.⁴ This reaction is likely to create interface defects such as vacancies, which will possess gap states.^{5,6} The Fermi level becomes pinned by the gap states of these interface defects. This model was supported by the observation that the energy of such defect states varied with composition in a similar way to the SBH and because the pinning occurs even for monolayer coverage of metal, before the MIGS would be established.

It is now believed that the intrinsic model gives a better description of Schottky barriers in most cases, because intrinsic states tend to give a larger pinning dipole, $N\delta$, than surface defects, as their density is higher (about one per surface atom) and they penetrate deeper into the semiconductor.² The apparent compatibility of the defect model arose because the CNL and the defect levels have a rather similar composition dependence.³³

The slope parameter S has been influential in our empirical understanding of Schottky barriers. Kurtin *et al.*³⁴ noted that S appeared to vary sharply with the ionicity of semiconductor (Fig. 2), from near 0 for low ionicity semiconductors like Si and GaAs to 1 for higher ionicity solids like SiO_2 , SrTiO_3 , and KTaO_3 . S is a dimensionless slope of barrier height to metal work function,

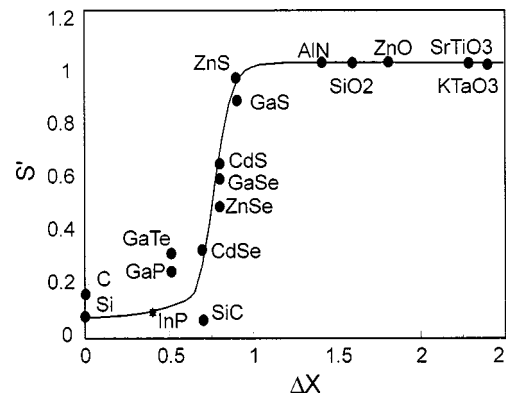


FIG. 2. Apparent Schottky barrier pinning factor S' vs semiconductor electronegativity difference ΔX , in the (incorrect) model of Kurtin *et al.* (see Ref. 31).

$$S = \frac{\partial \phi_n}{\partial \phi_M} \quad (4)$$

Louie *et al.*³⁵ and Schluter³⁶ noted that Kurtin *et al.*³⁴ had actually correlated the barrier heights to S' ,

$$S' = \frac{\partial \phi_n}{\partial X}, \quad (5)$$

which is the slope of barrier height to the Pauling electronegativity of the metal, and not the dimensionless S in Eq. (4). The work function and electronegativity vary roughly linearly as³⁷

$$\Phi_M = 2.27X_M + 0.34. \quad (6)$$

[For more recent tabulations,³⁸ the correlation is $\Phi_M = 1.79X_M + 1.11$ (Ref. 2).] Thus, $S' = 2.27S$, and the Schottky limit should correspond to $S' = 2.27$. The data rarely reach this limit and Schluter³⁶ observed that S had a better correlation with the dielectric constant of the semiconductor ϵ_0 . Later, Tersoff³¹ proposed a correlation with the electronic part of the dielectric constant, ϵ_∞ . Empirically, Mönch^{2,39} found that S obeyed

$$S = \frac{1}{1 + 0.1(\epsilon_\infty - 1)^2} \quad (7)$$

as seen in Fig. 3. Mönch⁴⁰ suggested that the dependence on ϵ_∞ arises because calculations of the $N\delta$ product in Eq. (3) scaled with $(\epsilon_\infty - 1)^{1.9}$, close to the power 2 required.

Certain materials are key tests of Schottky barrier models. Diamond^{41,42} and xenon⁴³ have zero ionicity but small ϵ_∞ , so their large values of S show that S depends on ϵ , not on ionicity.² This is tested by the plot $\log(S^{-1} - 1)$ vs $\log(\epsilon_\infty - 1)$ as in Fig. 4.

The wide gap oxides provide another important test.⁴⁴ SrTiO_3 and KTaO_3 were placed as high ionicity solids in the original Kurtin plot, with $S' \sim 1$. However, this was before data was actually known. When data⁴⁵ became available for SrTiO_3 showing S lying between 0.25 and 0.4 (Fig. 5), it was clear that S was much lower. The reason for this is that the SBHs depend on ϵ_∞ . ϵ_∞ is controlled by the states closest to

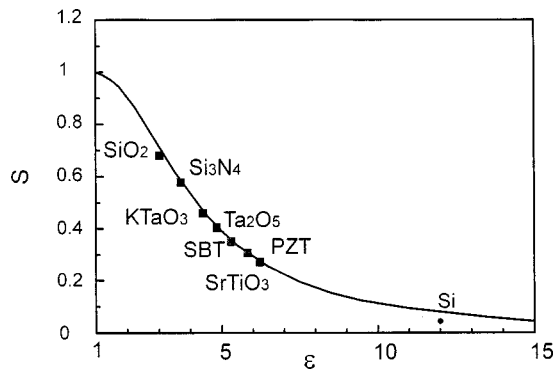


FIG. 3. Variation of the Schottky barrier S factor with electronic dielectric constant ϵ_{∞} from Eq. (7).

the band gap. In SrTiO_3 and KTaO_3 , these are Ti–O or Ta–O states of Ti–O or Ta–O bonds, which have a moderate ionicity. In contrast, the Sr–O or K–O bonds have a very high ionicity and their states lie well away from the gap and they provide a much smaller contribution to ϵ_{∞} . This can be seen in the partial density of states (DOS) of SrTiO_3 shown in Fig. 6. Thus, SrTiO_3 and KTaO_3 were misplaced on the original Kurtin plot (Fig. 2) as highly ionic solids. A lesser point is that the moderate value of S of SrTiO_3 clearly correlates with ϵ_{∞} , and not with the low frequency dielectric constant ϵ_0 , which has a very large value for ferroelectrics and would give $S \approx 0$ from Eq. (7).

In summary, the MIGS model of Schottky barrier heights holds for a wide range of solids, with a wide range of ionicities and dielectric constants. The layer compound GaSe is perhaps a remaining system in which the behavior seems to be still best explained in terms of interfacial reactions.⁴⁶

III. CHARGE NEUTRALITY LEVELS

We now consider the charge neutrality levels. The CNL is the branch point of the semiconductor interface states. It can be calculated from the Greens function of the band structure, taken over the Brillouin zone,²⁹

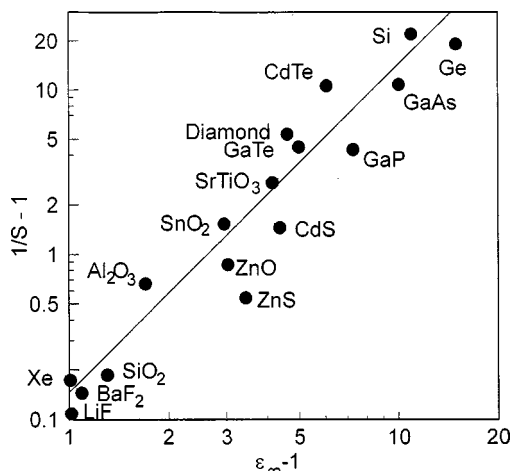


FIG. 4. Variation of Schottky barrier pinning factor S with electronic dielectric constant ϵ_{∞} after Ref. 2, with new data point for SrTiO_3 .

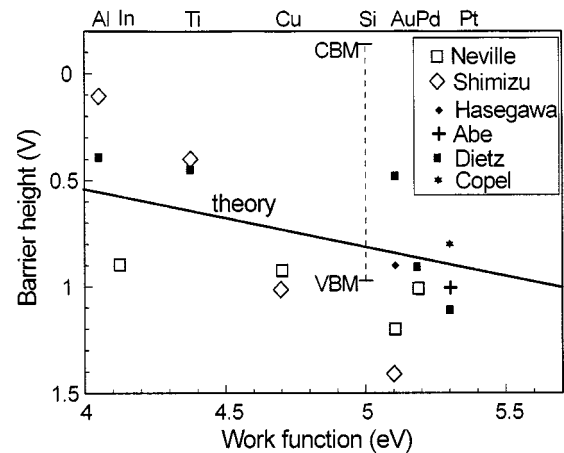


FIG. 5. Comparison of the predicted and experimental Schottky barrier heights for various metals on SrTiO_3 . For Si, the predicted band edges are shown.

$$G(E) = \int_{\text{BZ}} \frac{N(E') d'E}{E - E'} = 0. \quad (8)$$

Cardona and Christensen later provided a quicker method using a sum over special points of the Brillouin zone.^{47,48}

$$G(E) = \sum_i \frac{1}{E - E_i}. \quad (9)$$

The CNL is a weighted average of the valence and conduction band DOS.

$$E_{\text{CNL}} = \frac{N_1 E_2 + N_2 E_1}{N_1 + N_2}. \quad (10)$$

The CNL is the balance point of the weights of the valence and conduction band DOSs. Thus, a high density of states in the valence band tends to push the CNL towards the conduction band and vice versa.

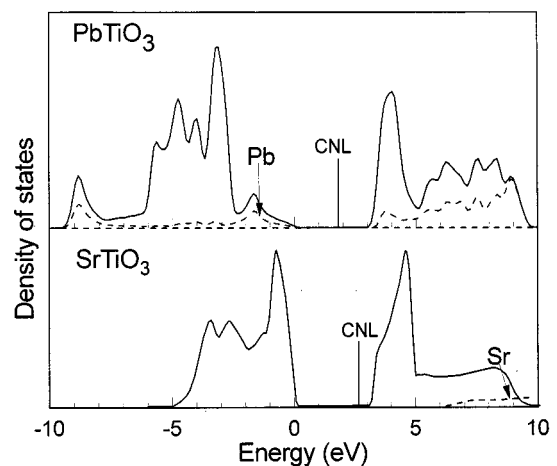


FIG. 6. Partial density of states of (a) PbTiO_3 and (b) SrTiO_3 . (b) shows that the Ti–O states lie close to the band gap, while the Sr states lie high in the conduction band. (b) shows that Pb s and p states lie close to the band edge in PbTiO_3 , which tends to lower the CNL compared to SrTiO_3 .

The CNLs show interesting chemical trends. The CNLs have been tabulated by Mönch⁴⁸ for most diamond, zinc-blende, and wurzite semiconductors. The CNL lies at the center of the average bonding-antibonding gap, rather than of the minimum direct or indirect gap. The CNL lies in the lower part of the gap for materials with the low ionicity such as C, Si, or GaAs. This accounts for one of the first rules of thumb, that the Schottky barrier height for holes equals 1/3 of the band gap.⁴⁹ The CNL falls lower in the gap along the sequence C to Ge, because overall the conduction bands spread over a smaller energy range. As a function of ionicity or polarity, the CNL tends to rise in the gap. This is because the upper valence bands become flatter (higher effective mass) with increasing polarity, while the conduction bands have increasingly direct gap character, so they have a relatively lower DOS towards the band edge. Note that the nitrides stand out as having a higher polarity than other III–V solids, as is apparent in their other properties.

IV. SEMICONDUCTOR BAND ALIGNMENTS

The band alignment between two semiconductors depends on whether or not there is an interface dipoles, just as in the case of Schottky barriers. In the limit of no charge transfer dipoles, equivalent to the Schottky limit, the conduction band offsets is given by the difference in their electron affinities. This is known as the electron affinity model.⁵⁰ A similar idea was that in the absence of charge transfer, the band lineups can be derived by placing the band energies of each semiconductor on an absolute energy scale. One such scale is the vacuum level of the free atom energy levels in the tight-binding bands.⁵¹

Tersoff^{31,32} showed that the band alignment between two semiconductors *a* and *b* is controlled by the charge transfer across the interface and the resulting interface dipoles, in a similar fashion to the Schottky barrier. Charge transfer across the interfacial bonds creates a dipole, which modifies the band lineup given by the electron affinity rule,

$$\phi_n = (\chi_a - \Phi_{\text{CNL},a}) - (\chi_b - \Phi_{\text{CNL},b}) + S(\Phi_{\text{CNL},a} - \Phi_{\text{CNL},b}). \quad (11)$$

The lineup is now described by a matching of the charge neutrality levels of each semiconductor, modified by the *S* factor. For simple semiconductors like Si, ϵ_∞ is large so *S* is small and the third term was negligible in the original formulation, but it is retained here for wide gap semiconductors. For strong pinning, the alignment is just given by the alignment of the two CNLs. The CNL energy below the vacuum level is a measure of the mean electronegativity of the semiconductor, in the same way that the work function of a metal is proportional to the metal's electronegativity. Thus, Eq. (11) says that the band alignment is the difference in electronegativity, screened by the *S* factor.

A wide ranging quantitative comparison⁸ has found that the general class of charge neutrality models gives a good description of the band offsets, and that the numerical estimates from the Cardona–Christensen⁴⁷ method are as reliable as those from more complicated methods. A notable

TABLE I. Experimental band gaps, dielectric constants ϵ_∞ , and electron affinity (EA), and calculated Schottky barrier slopes *S*, charge neutrality level above the valence band edge (CNL), and conduction band offset on Si for each oxide.* is estimated value.

	Gap (eV)	CNL (eV)	ϵ_∞	<i>S</i> calculated	EA (eV)	Calculated CB offset (eV)
Si	1.1	0.2	12		4.0	
SiO ₂	9		2.25	0.86	0.9	3.5 (exp)
Si ₃ N ₄	5.3		3.8	0.51	2.1	2.4 (exp)
SrTiO ₃	3.3	2.60	6.1	0.28	3.9	−0.1
PbTiO ₃	3.4	1.90	6.25	0.31	3.5	0.6
BaZrO ₃	5.3	3.7	4	0.53	2.5	0.8
PbZrO ₃	3.7	2.6	4.8	0.4	3.2	0.2
Ta ₂ O ₅	4.4	3.3	4.84	0.40	3.2	0.36
SrBi ₂ Ta ₂ O ₉	4.1	3.3	5.3	0.35	3.3	0.15
TiO ₂	3.05	2.2	7.8	0.18	3.9*	0
ZrO ₂	5.8	3.6	4.8	0.41	2.5*	1.4
HfO ₂	6	3.7	4	0.53	2.5*	1.5
Al ₂ O ₃	8.8	5.5	3.4	0.63	1*	2.8
Y ₂ O ₃	6	2.4	4.4	0.46	2*	2.3
La ₂ O ₃	6*	2.4	4	0.53	2*	2.3
ZrSiO ₄	6*	3.6	3.8	0.56	2.5*	1.5
HfSiO ₄	6*	3.6	3.8	0.56	2.5*	1.5

feature is that a common energy scale, as in the zero charge transfer limit, leads to small valence offsets for compounds with the same anion, “the common anion rule,”⁵² whereas the alignment of CNLs gives quite large valence offsets in these cases, which is the observed situation.³⁰

V. APPLICATION TO OXIDES

The band offsets for the various wide gap oxides in contact with metal or silicon are found by calculating their *S* parameter and their CNLs. The *S* factors are found from Eq. (7) using the experimental values of ϵ_∞ and are shown in Table I. The CNLs are found by calculating the oxide band structures by the tight-binding method as previously.^{44,53–56} The tight-binding parameters are found by scaling,⁵⁷ by fitting to existing band structures,^{58,59} and by fitting to photoemission spectra and optical data.^{60–67} Tight binding generally describes well the valence bands of *s,p* bonded semiconductors but is less successful for the less localized conduction band states. However, in the transition metal oxides studied here, the conduction bands are formed from localized *d* states, which can be successfully described by tight binding.⁵⁸ The CNLs are found as the zeros of the Green's function of the bulk band structures, determined at the special points of the Brillouin zone.

The charge neutrality levels for the various oxides are given in Table I together with the experimental values of their band gaps and electron affinities.^{62–67} In Eq. (11), Φ_{CNL} is the energy of the CNL below the vacuum level; it is given in Table I as ϕ_{CNL} its energy above the valence band edge.

It should be noted that the band alignments for narrower gap oxides with $S=0.2$ – 0.3 like SrTiO₃ depend mainly on their charge neutrality levels, whereas the alignments for wider gap oxides with $S>0.5$ depend more strongly on their electron affinity. The EAs of many oxides are not known, but

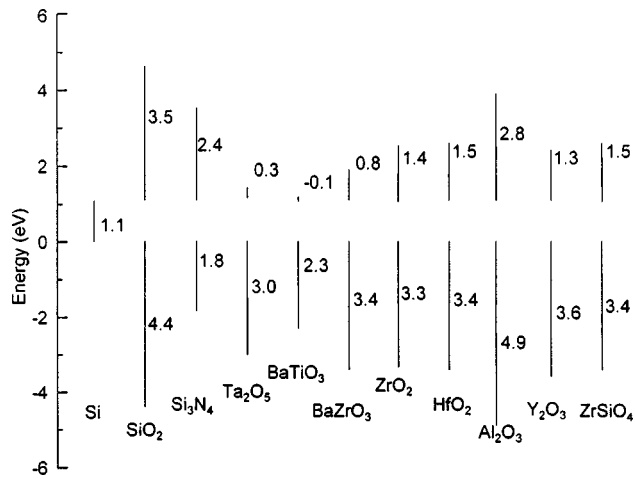


FIG. 7. Calculated conduction band and valence band offsets of various oxides on Si.

they can be a factor leading to lower CB offsets. This emphasises the need for more photoemission measurements on these systems to determine the photoelectric thresholds and electron affinities.

The calculated barrier heights of the oxides are given in Table I and shown schematically in Fig. 7. We see that the band offsets for most transition metal oxides are quite asymmetric, with barriers for electrons much less than half the oxide band gap. For Ta_2O_5 , the electron barrier on Pt is 1.42 eV and only 0.36 eV for Ta_2O_5 on Si. Conduction data for Ta_2O_5 films show evidence of both Schottky emission over a barrier and Poole–Frenkel tunneling, but with no estimate of the barrier height. Indeed, there have been no direct photoemission measurements of barrier heights on Ta_2O_5 , a key oxide. Our calculations show that Schottky emission of electrons into the conduction band will be a significant cause of leakage for Ta_2O_5 films on Si. On the other hand, the barrier for holes is quite large, 2.9 eV. The small barrier for electrons on Si should make Ta_2O_5 a poor gate oxide. Indeed Chatterjee *et al.*¹⁶ have found a relatively small activation energy for leakage in Ta_2O_5 . Nevertheless, working FETs with Ta_2O_5 gate oxide have been made,¹⁴ suggesting that leakage can be limited. A problem with Ta_2O_5 is that it is thermodynamically unstable in contact with Si.⁶⁸

Figure 5 compares the predicted Schottky barrier heights of SrTiO_3 on various metals with the experimental values.^{45,17,69–72} SrTiO_3 is the most studied system and the best test of our calculations. The experimental data are quite scattered but are quite consistent with $S \ll 1$ and our calculated value of 0.28. This shows that SrTiO_3 is a key oxide in the tests of Schottky barrier models. Note that we cannot account for the much larger S value found in by Shimizu *et al.*⁷¹ In detail, the calculated barrier height for SrTiO_3 on Pt is 0.9 eV which is close to 0.8 eV found by photoemission by Copel *et al.*⁷² and 1.1 eV found by Dietz *et al.*¹⁷ for BST. Photoemission is perhaps the most reliable estimate. The low barriers found here support the observation that Schottky emission into the conduction band is a significant source of leakage in BST films.^{17,18}

A serious result is that the calculated conduction band offset for BST on Si is actually -0.1 eV, that there is no barrier,^{43,56} as seen in Fig. 5. At present, there are no data to confirm this prediction. The BST:Si interface is reactive and a special synthesis sequence must be used.⁷³ Generally, Ir/Pt diffusion barriers are interposed between BST and poly-Si in DRAM, so there is no direct contact.

$\text{Pb}(\text{Ti}_x\text{Zr}_{1-x}\text{O}_3$ with $x \approx 0.55$) or PZT is an important ferroelectric for nonvolatile memories and other applications. The predicted barrier height for Pt on PZT ($\text{Pb}_{0.55}\text{Zr}_{0.45}\text{O}_3$) is 1.45 eV, which is close to the 1.5 eV value measured by Dey *et al.*⁷⁴ The larger value of the hole barrier than the electron barrier means that PZT thin films can have predominantly electron injection, even though bulk PZT tends to be p type. This contrasts with the model of Wouters *et al.*⁷⁵ who proposed hole injection. The electron barrier of Pt on PZT is larger than on BST because the CNL in PZT lies lower in the gap. This is because of the different band structure of PZT, in which the Pb 6s and 6p states form the band edges and this tends to lower the CNL.

$\text{SrBi}_2\text{Ta}_2\text{O}_9$ (SBT) is an important ferroelectric for nonvolatile memories.^{21,22,63} Note that more recent optical data^{61–63} find that the band gap of SBT is 4.1 eV, not 5.3 eV as originally.⁵⁴ The Schottky barrier of Pt is predicted to be 1.2 eV, which is essentially the same as found by photoemission.⁷⁶ SBT is similar to PbTiO_3 in that the Bi s and p states form both band edges,⁵⁴ and this tends to lower the CNL in the gap. However, the Bi stoichiometry is less, so the effect is not as large.

We now consider the band alignments of alternative gate dielectrics. The Roadmap of the Semiconductor Industries Association proposes Ta_2O_5 as the high-K dielectric for the next generation devices and BST for the second generation. Considerable work has already been carried out on these two insulators.^{9–25,68} TiO_2 , Al_2O_3 , ZrO_2 , HfSiO_4 , and ZrSiO_4 are all being presently studied as alternative dielectrics.^{26,77–82}

Figure 7 shows the calculated band offsets of the oxides on Si. The experimental band offsets of SiO_2 and Si_3N_4 on Si are included for completeness.⁸³ A band offset of over about 1 eV is needed to keep Schottky emission low enough. We see that the SrTiO_3 and Ta_2O_5 have too low conduction band (CB) offsets with Si, as already mentioned. This is fundamentally because their CNL lies asymmetrically in the upper part of the band gap, so the electron barrier is small while the hole barrier is larger than necessary. It would be possible to raise the electron barrier if the band alignments could be more symmetric.

To increase the CB offset, it is necessary to increase the band gap and/or lower the CNL. The oxides of interest are transition metal (TM) oxides, in which the band gap is between the valence band of filled oxygen 2p states and the conduction band of empty transition metal d states. The band gap can be increased by raising the TM d levels. A tabulation of atomic energy levels⁵⁷ shows that this can be achieved by using 4d or 5d metals instead of 3d metals, or using transi-

tion metals from column III instead of column IV of the Periodic Table.

Following this strategy, zirconates have wider gaps than titanates. Barium zirconate BaZrO_3 has a 2 eV wider gap than BaTiO_3 . Figure 6 shows that its predicted CB offset is raised to 0.8 eV, but this is still below the desired 1 eV.

An insulator with very wide gap is Al_2O_3 . Its gap is 8.8 eV,⁶⁴ and its band structure is well known. Table I shows that its valence band (VB) and CB offsets are large, and it is acceptable. However, Al_2O_3 represents a ‘‘last resort’’ in that its static dielectric constant is quite small. Although the permittivity of Al_2O_3 is sufficiently large to give the capacitance density of 10–15 $\text{\AA}$ SiO_2 with a physical thickness large enough to give a reasonable tunneling barrier, it is only likely to last for one device generation. That is, further device scaling would be thwarted by the same problems presently associated with SiO_2 .

A better strategy is to lower the CNL within the gap. Equation (10) showed that the CNL is a weighted average of the density of states (DOS). The CNL is higher if the VB DOS is larger than the CB DOS. The oxides of interest have relatively ionic bonding, so their valence bands consist mainly of oxygen (anion) p states and their conduction bands consist mainly of metal (cation) d states. Thus, the ratio of VB DOS to CB DOS is essentially the stoichiometry ratio of anions to cations. The CNL is lowered if the metal valence is lowered from 4 to 3. We see that the CNL is much lower in the gap in Y_2O_3 , La_2O_3 , and Al_2O_3 . Y_2O_3 and La_2O_3 are the oxides with largest CB offsets for reasonable dielectric constants.

Another way to reduce the anion to cation stoichiometry is to use the oxide BO_2 rather than the perovskite ABO_3 . ZrO_2 has recently been studied by Copel *et al.*⁷⁸ Its band gap of 5.8 eV (Ref. 65) is slightly wider than that of BaZrO_3 . This and the stoichiometry difference give a larger predicted CB offset for ZrO_2 (1.4 eV) than BaZrO_3 , and indeed one which is just sufficiently high.

There are considerable advantages in using amorphous dielectrics. These have no grain boundaries or surface facets, which could cause higher leakage currents or low local permittivity. They also avoid problems with grain sizes, which lead to irreproducibility when they are of the order of device sizes. The alternative is to use a perfectly lattice matched, epitaxial oxide layer.⁷³ ZrO_2 has an amorphous phase if process temperatures are less than 700 °C.⁷⁸ A disadvantage of Y_2O_3 , ZrO_2 , and other isostructural oxides such as HfO_2 and CeO_2 is that they are fast ion conductors with a significant oxygen ion diffusivity.

Zirconium silicate (ZrSiO_4) is an amorphous Zr compound in which the oxygen atoms are covalently bonded, and so its oxygen ions are strongly bound. The structure of ZrSiO_4 consists of chains of alternate edge-sharing ZrO_4 and SiO_2 tetrahedra, with additional Zr–O bonds between the chains, leading to an overall sixfold Zr coordination.⁸² The edge-sharing tetrahedra give four-atom Zr–O–Si–O rings. Wilk *et al.*^{80,81} have recently found that the leakage currents of HfSiO_4 and ZrSiO_4 films to be very good. We estimate

the band gap of ZrSiO_4 to be 6 eV, similar to ZrO_2 , but there are no experimental values to the author’s knowledge. An estimated electron affinity is also used. Using these values, the CB offset is predicted to be 1.5 eV, slightly larger than for ZrO_2 . This accounts for its low leakage currents. An advantage of ZrSiO_4 and HfSiO_4 technologically is that they can be introduced as the alloy $(\text{SiO}_2)_{1-x}(\text{ZrO}_2)_x$, starting at low x and increasing x as experience is developed. A drawback to ZrSiO_4 is its bulk dielectric constant of ~ 11 . Although this compound may represent the suitable material for future gate insulators, the ability to produce multiple device generations is unclear.

These results show that Ta_2O_5 and BST, the oxides on which most effort has so far been expended, are not suitable for gate oxides because of their low electron barrier heights. The most suitable oxides are those with lower K , such as ZrO_2 and ZrSiO_4 . Are there any ‘‘fix ups’’ that can be played to allow Ta_2O_5 and BST to be used? The main problem for Ta_2O_5 is the asymmetry of the band alignment. The introduction of an interface dipole to raise the oxide bands with respect to Si would fix the problem. A second alternative is to allow a monolayer of SiO_2 at the interface.⁸² The large S of SiO_2 would unscreen the interface, raise S and stop the charge transfer. The alignment now reverts to a difference in electron affinities. The EAs in Table I suggest that the conduction band alignment increases from 0.35 to 0.8 eV. Similarly, the epitaxial SrTiO_3 gate oxides on Si have thin BaO or SiO_2 interface layers^{73,84} and this may allow them to have larger electron barriers. However, note that the predictions in the screened limit are reasonably safe because they depend on the CNLs which depend only on the band structure. In the $S=1$ limit, the alignment depends only on the EAs, and these are not known for many oxides in Table I. This emphasises the need for photoemission data.

In conclusion, the theory of band lineups of semiconductors was used to predict Schottky barrier heights and band offsets of Ta_2O_5 , SrTiO_3 , PbTiO_3 , and $\text{SrBi}_2\text{Ta}_2\text{O}_9$ on various metals and on Si. In each case, on Si the barrier for electrons is much less than for holes, and it suggests that electron Schottky emission is likely to be a significant source of leakage in these films. ZrO_2 and ZrSiO_4 are found to have sufficiently large electron barriers for use as gate oxides.

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¹E. H. Rhoderick and R. H. Williams, *Metal Semiconductor Contacts* (Oxford University, Oxford, 1988).

²W. Mönch, *Surf. Sci.* **300**, 928 (1994).

³J. Tersoff, *Surf. Sci.* **168**, 275 (1986).

⁴L. J. Brillson, *Surf. Sci.* **300**, 909 (1994).

- ⁵W. E. Spicer, T. Kendelewicz, N. Newman, K. K. Chin, and I. Lindau, *Surf. Sci.* **168**, 240 (1986).
- ⁶R. E. Allen, O. F. Sankey, and J. D. Dow, *Surf. Sci.* **168**, 376 (1986).
- ⁷A. D. Katnani and G. Margaritondo, *Phys. Rev. B* **28**, 1944 (1983).
- ⁸E. T. Yu, J. O. McCaldin, and T. C. McGill, *Solid State Phys.* **46**, 1 (1992).
- ⁹K. W. Kwon, C. S. Kang, S. O. Park, H. K. Kang, and S. T. Ahn, *IEEE Trans. Electron Devices* **43**, 919 (1996).
- ¹⁰R. A. B. Devine, L. Vallier, J. L. Autran, P. Paillet, and J. L. Leray, *Appl. Phys. Lett.* **68**, 1775 (1996).
- ¹¹Y. Matsui *et al.*, *IEEE Electron Device Lett.* **17**, 431 (1996).
- ¹²G. B. Alers, R. M. Fleming, Y. H. Wong, B. Dennis, A. Pinczuk, G. Redinbo, R. Urdahl, E. Ong, and Z. Hasan, *Appl. Phys. Lett.* **72**, 1308 (1998).
- ¹³S. Ezhilvalavan and T. Y. Tseng, *J. Appl. Phys.* **83**, 4797 (1998).
- ¹⁴D. Park, Y. King, T. J. King, C. Hu, A. Kalnitsky, S. P. Tay, and C. C. Cheng, *IEEE Electron Device Lett.* **19**, 441 (1998).
- ¹⁵R. Roy and I. C. Kizilyalli, *Appl. Phys. Lett.* **72**, 2835 (1998).
- ¹⁶A. Chatterjee *et al.*, *IEDM Tech. Digest*, 777 (1998).
- ¹⁷G. W. Dietz, W. Antpohler, M. Klee, and R. Waser, *J. Appl. Phys.* **78**, 6113 (1995).
- ¹⁸G. W. Dietz, M. Schumacher, R. Waser, S. K. Streiffer, C. Basceri, and A. I. Kingon, *J. Appl. Phys.* **82**, 2359 (1997).
- ¹⁹D. E. Kotecki, *Integr. Ferroelectr.* **16**, 1 (1997).
- ²⁰C. Sudhama, A. C. Campbell, P. D. Maniar, R. E. Jones, and C. J. Mogab, *J. Appl. Phys.* **75**, 1014 (1994).
- ²¹C. A. Paz de Araujo, J. D. Cuchiaro, L. D. McMillan, M. C. Scott, and J. F. Scott, *Nature (London)* **374**, 627 (1995).
- ²²J. F. Scott, *Annu. Rev. Mater. Sci.* **28**, 79 (1998).
- ²³G. Lucovsky, H. Y. Yang, Y. Wu, and H. Niimi, *IEEE Electron Device Lett.* (in press).
- ²⁴P. Zurcher *et al.*, *Mater. Res. Soc. Symp. Proc.* **541**, 11 (1999).
- ²⁵A. Grill *et al.*, *Mater. Res. Soc. Symp. Proc.* **541**, 89 (1999).
- ²⁶H. J. Hubbard and D. G. Schlom, *J. Mater. Res.* **11**, 2757 (1996).
- ²⁷A. W. Cowley and S. M. Sze, *J. Appl. Phys.* **36**, 3212 (1965).
- ²⁸V. Heine, *Phys. Rev.* **138A**, 1689 (1965).
- ²⁹J. Tersoff, *Phys. Rev. Lett.* **52**, 465 (1984).
- ³⁰J. Tersoff, *Phys. Rev. Lett.* **56**, 2755 (1986).
- ³¹J. Tersoff, *Phys. Rev. B* **30**, 4874 (1984).
- ³²J. Tersoff, *Phys. Rev. B* **32**, 6968 (1985).
- ³³J. Tersoff and W. A. Harrison, *Phys. Rev. Lett.* **58**, 2367 (1987).
- ³⁴S. Kurtin, T. C. McGill, and C. A. Mead, *Phys. Rev. Lett.* **30**, 1433 (1969).
- ³⁵S. G. Louie, J. R. Chelikowsky, and M. L. Cohen, *Phys. Rev. B* **15**, 2154 (1977).
- ³⁶M. Schluter, *Phys. Rev. B* **17**, 5044 (1978); *Thin Solid Films* **93**, 3 (1982).
- ³⁷W. Gordy and W. J. O. Thomas, *Phys. Rev.* **24**, 439 (1956).
- ³⁸H. B. Michaelson, *J. Appl. Phys.* **48**, 4729 (1977).
- ³⁹W. Mönch, *Phys. Rev. Lett.* **58**, 1260 (1986).
- ⁴⁰W. Mönch, *Appl. Surf. Sci.* **92**, 367 (1996).
- ⁴¹W. Mönch, *Europhys. Lett.* **27**, 479 (1994).
- ⁴²M. Aoki and H. Kawarada, *Jpn. J. Appl. Phys., Part 2* **33**, L708 (1994).
- ⁴³W. Jacob, E. Bertel, and V. Dose, *Europhys. Lett.* **4**, 1303 (1987).
- ⁴⁴J. Robertson and C. W. Chen, *Appl. Phys. Lett.* **74**, 1168 (1999).
- ⁴⁵R. C. Neville and C. A. Mead, *J. Appl. Phys.* **43**, 4657 (1972).
- ⁴⁶R. H. Williams, A. McKinley, G. J. Hughes, V. Montgomery, and I. T. McGovern, *J. Vac. Sci. Technol.* **21**, 594 (1982).
- ⁴⁷M. Cardona and N. E. Christensen, *Phys. Rev. B* **35**, 6182 (1987).
- ⁴⁸W. Mönch, *J. Appl. Phys.* **80**, 5076 (1996).
- ⁴⁹C. A. Mead and W. G. Spitzer, *Phys. Rev.* **134A**, 713 (1964).
- ⁵⁰R. L. Andersen, *Solid-State Electron.* **5**, 341 (1962).
- ⁵¹W. A. Harrison, *J. Vac. Sci. Technol.* **14**, 1016 (1977).
- ⁵²J. O. McCaldin, T. C. McGill, and C. A. Mead, *Phys. Rev. Lett.* **36**, 56 (1976).
- ⁵³J. Robertson, W. L. Warren, B. A. Tuttle, D. Dimos, and D. M. Smyth, *Appl. Phys. Lett.* **63**, 1519 (1993).
- ⁵⁴J. Robertson, C. W. Chen, W. L. Warren, and C. D. Gutleben, *Appl. Phys. Lett.* **69**, 1704 (1996).
- ⁵⁵J. Robertson, C. W. Chen, and W. L. Warren, *Mater. Res. Soc. Symp. Proc.* **493**, 249 (1998).
- ⁵⁶J. Robertson, E. Raissi, J. P. Maria, and A. I. Kingon, *Mater. Res. Soc. Symp. Proc.* **592** (1999).
- ⁵⁷W. A. Harrison, *Electronic Structure* (Freeman, San Francisco, 1979).
- ⁵⁸L. F. Mattheis, *Phys. Rev. B* **6**, 4718 (1972).
- ⁵⁹R. D. King-Smith and D. Vanderbilt, *Phys. Rev. B* **49**, 5828 (1994).
- ⁶⁰G. S. Oehrlein, *J. Appl. Phys.* **55**, 3715 (1984).
- ⁶¹C. H. Seager (private communication).
- ⁶²A. J. Hartmann, R. M. Lamb, J. F. Scott, P. N. Johnston, M. El Bouanani, C. W. Chen, and J. Robertson, *J. Korean Phys. Soc.* **32**, S1329 (1998).
- ⁶³J. F. Scott, *Ferroelectr. Rev.* **1**, 1 (1998).
- ⁶⁴R. H. French, *J. Am. Ceram. Soc.* **73**, 477 (1990).
- ⁶⁵R. H. French, S. J. Glass, F. S. Ohuchi, Y. N. Xu, and W. Y. Ching, *Phys. Rev. B* **49**, 5133 (1994).
- ⁶⁶J. Pascual, J. Camassel, and H. Mathieu, *Phys. Rev. B* **18**, 5606 (1978).
- ⁶⁷E. D. Palik, *Handbook of Optical Properties of Solids* (Academic, New York, 1985), Vols. 1–3.
- ⁶⁸G. B. Alers, D. J. Werder, Y. Chabal, H. C. Lu, E. P. Gusev, E. Garfunkel, T. Gustafsson, and R. S. Urdahl, *Appl. Phys. Lett.* **73**, 1517 (1998).
- ⁶⁹H. Hasegawa and T. Nishino, *J. Appl. Phys.* **69**, 1501 (1991).
- ⁷⁰K. Abe and S. Komatsu, *Jpn. J. Appl. Phys., Part 1* **31**, 2985 (1992).
- ⁷¹T. Shimizu, N. Gotoh, N. Shinozaki, and H. Okushi, *Appl. Surf. Sci.* **117**, 400 (1997).
- ⁷²M. Copel, P. R. Duncombe, D. A. Neumayer, T. M. Shaw, and R. M. Tromp, *Appl. Phys. Lett.* **70**, 3227 (1997).
- ⁷³R. A. McKee, F. J. Walker, and M. F. Chisholm, *Phys. Rev. Lett.* **81**, 3014 (1998).
- ⁷⁴S. K. Dey, J. J. Lee, and P. Alluri, *Jpn. J. Appl. Phys., Part 1* **34**, 3134 (1995).
- ⁷⁵D. J. Wouters, G. J. Willems, and H. E. Maes, *Microelectron. Eng.* **29**, 249 (1995).
- ⁷⁶C. D. Gutleben, *Appl. Phys. Lett.* **71**, 3444 (1997).
- ⁷⁷L. Manchanda *et al.*, *Tech. Dig. IEDM*, 605 (1998).
- ⁷⁸M. Copel, M. Gribelyuk, and E. Gusev, *Appl. Phys. Lett.* **76**, 436 (2000).
- ⁷⁹H. S. Kim, D. C. Gilmer, S. A. Campbell, and D. L. Polla, *Appl. Phys. Lett.* **69**, 3860 (1996).
- ⁸⁰G. D. Wilk and R. M. Wallace, *Appl. Phys. Lett.* **74**, 2854 (1999).
- ⁸¹G. D. Wilk, R. M. Wallace, and J. M. Antony, *J. Appl. Phys.* **87**, 484 (2000).
- ⁸²G. Lucovsky (private communication).
- ⁸³J. W. Keister, J. E. Rowe, J. J. Kolodziej, H. Niiki, T. E. Madey, and G. Lucovsky, *J. Vac. Sci. Technol. B* **17**, 1831 (1999).
- ⁸⁴K. Eisenberger *et al.*, *Appl. Phys. Lett.* **76**, 1324 (2000).